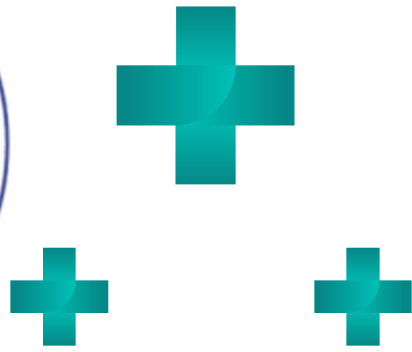




WELCOME

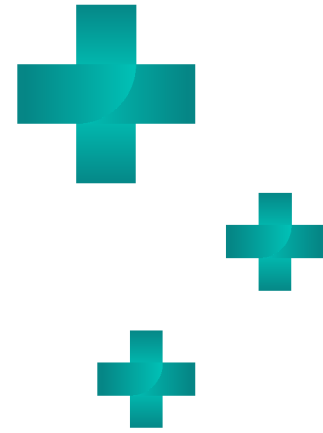


TO OUR PROJECT PRESENTATION

On

Hospital Management System

Presented To



Presented By



K.M. Tahlil Mahfuz Faruk
Lecturer

**Department of Software
Engineering**

Green University of Bangladesh

NAME:Radha Nath Sarker (Bishal)

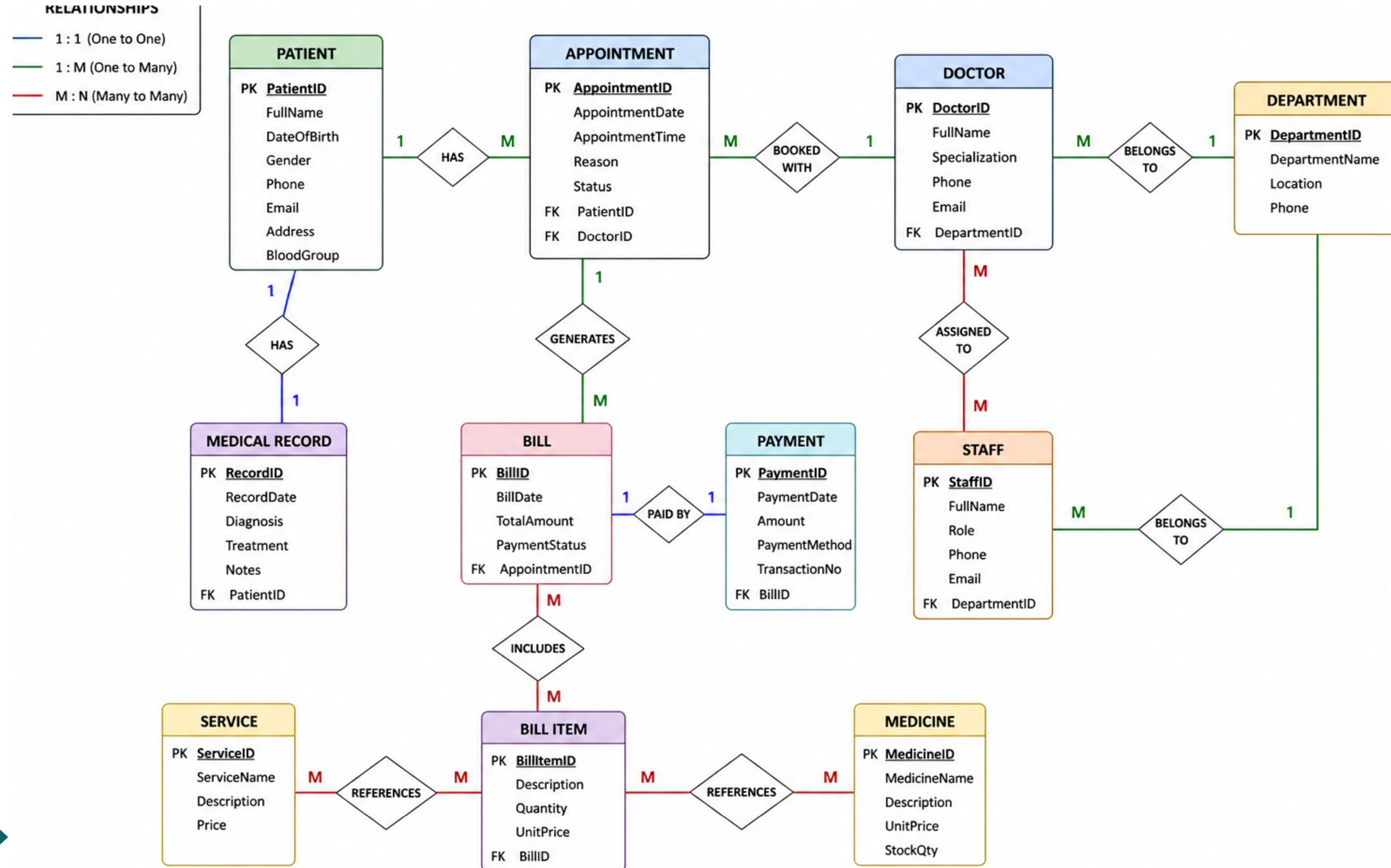
ID:242034014

Department:Software Engineering

01/05/2026

02

ER Diagram :Hospital Managment System



NORMALIZATION



Stage	Description
UNF	Raw data with repeating groups
1NF	Atomic values, no repeating groups
2NF	No partial dependency
3NF	No transitive dependency

SCHEMA + CONSTRAINTS

Student Table

Column	Type	Constraints
stu_id	INT	PK, NOT NULL
email	VARCHAR	UNIQUE
age	INT	CHECK (age ≥ 18)

Enroll (junction for M:N)

Column	Type	Constraints
stu_id	INT	FK → Student
course_id	INT	FK → Course

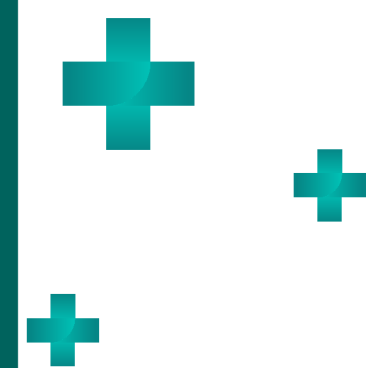


FUNCTION & STORED PROCEDURE



```
CREATE FUNCTION get_total(price DECIMAL, qty INT)
  RETURNS DECIMAL
  RETURN price * qty;
```

```
CREATE PROCEDURE new_order(IN cid INT, IN total DEC)
  BEGIN
  INSERT INTO Orders(cust_id, total) VALUES(cid, total);
  END;
```





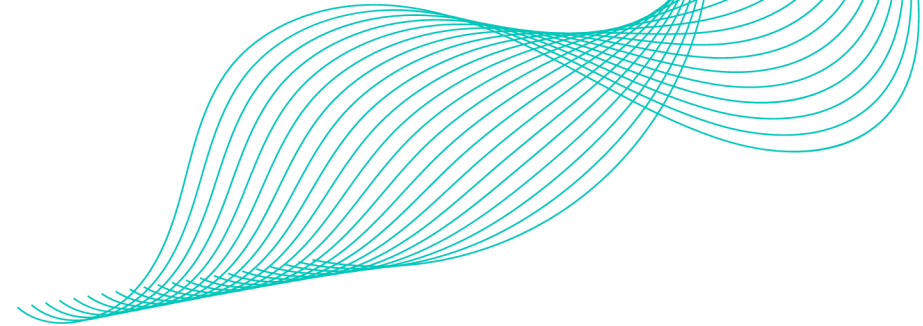
TRIGGER + CURSOR

Trigger

```
CREATE TRIGGER price_log  
AFTER UPDATE ON Products  
FOR EACH ROW  
INSERT INTO PriceLog(old_price, new_price)  
VALUES(OLD.price, NEW.price);
```

Cursor

```
DECLARE cur CURSOR FOR SELECT salary FROM Employees;
```



CARDINALITY PROOF

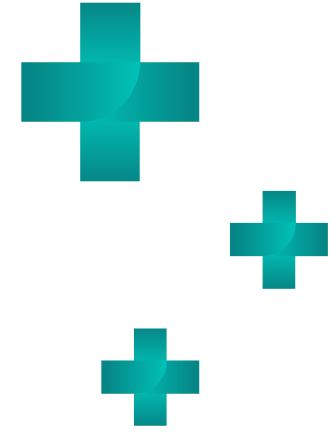
Type	Implementation	Proof
1:1	Users ↔ Passports (shared PK)	One user → one passport
1:M	Customer → Orders	FK in Orders table
M:N	Students ↔ Courses via Enrollments	Junction table with two FKs



CONCLUSION



**All requirements met: Normalization (3NF), Constraints, Function, Procedure,
Trigger, Cursor, ERD with cardinalities
Database is consistent, efficient, and ready for scaling**



THANK YOU